

4 (a)

$$a = r\omega^2$$

$$= 4.22 \times 10^8 \left(\frac{2\pi}{1.53 \times 10^5} \right)^2$$

$$= 0.712 \text{ m s}^{-2}$$

(b)(i)

Direction:

The centripetal force acting on each moon is due to the gravitational force of attraction by Jupiter on the moons. Hence the gravitational field strength has the same direction as the centripetal acceleration of the moon.

Magnitude:

Since gravitational force provides the centripetal force,

$$F_g = F_c$$

$$m_{\text{moon}} g_{\text{Jupiter}} = m_{\text{moon}} a_c$$

where $g_{Jupiter}$: Jupiter's gravitational field strength and

a_r : centripetal accleration of each moon.

Hence the gravitational field strength has the same magnitude as the centripetal acceleration of the moon.

(b)(ii)

$$\begin{aligned}\frac{a_{Io}}{a_{Amalthea}} &= \frac{r_{Amalthea}^2}{r_{Io}^2} \\ \frac{r_{Io}}{r_{Amalthea}} &= \sqrt{\frac{a_{Amalthea}}{a_{Io}}} \\ &= \sqrt{\frac{3.87}{0.7117}} \\ &= 2.33\end{aligned}$$

5 (a)(i)

$$F_A = q_A E$$